

FET

ELECTRONICS · SEMICONDUCTOR DEVICES

Field Effect Transistors

A comprehensive infographic guide covering JFET & MOSFET —
characteristics, parameters, biasing, and operating regions.

JFET

MOSFET

BIASING

PARAMETERS

OHMIC REGION

PRESS → TO BEGIN

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← → NAVIGATE

SECTION 01 · INTRODUCTION

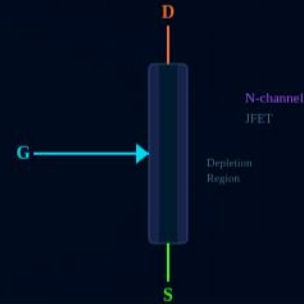
JFET – Junction Field Effect Transistor

What is a JFET?

A voltage-controlled semiconductor device where current flow between **Drain (D)** and **Source (S)** is controlled by a reverse-biased **Gate (G)** junction voltage, with no gate current required.

Key Advantage

Extremely **high input impedance** ($M\Omega$ range) — unlike BJTs, JFETs draw virtually no gate current, making them ideal for sensitive analog amplifiers.

**3**

Terminals

2

Types (N/P)

 ∞

Input Impedance

V

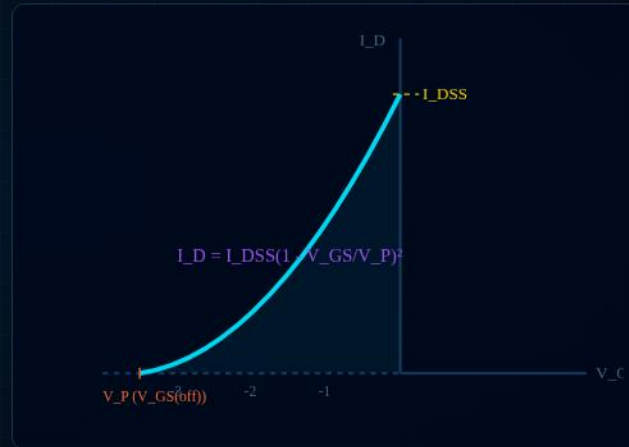
Voltage Controlled

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SECTION 02 • JFET

JFET Characteristics

DRAIN CHARACTERISTICS (I_D vs V_{DS})TRANSFER CHARACTERISTICS (I_D vs V_{GS})

SHOCKLEY EQUATION

$$I_D = I_{DSS} (1 - V_{GS}/V_P)^2$$

SECTION 03 · JFET

JFET Parameters

Symbol	Parameter	Typical
I_{DSS}	Drain-to-Source Saturation Current	1–20 mA
V_p	Pinch-off Voltage ($V_{GS(off)}$)	-2 to -10 V
g_m	Transconductance	1–8 mS
r_d	Drain Resistance	10–1000 kΩ
μ	Amplification Factor ($\mu = g_m r_d$)	10–1000
C_{gd}	Gate-Drain Capacitance	1–10 pF

Transconductance Formula

$$g_m = \Delta I_D / \Delta V_{GS}$$

$$g_m = g_{m0} (1 - V_{GS}/V_p)$$

where $g_{m0} = 2I_{DSS}/|V_p|$ is the maximum transconductance at $V_{GS} = 0$.

N-Channel vs P-Channel

Electrons	CARRIERS	Holes
$V_{GS} < 0$	BIAS	$V_{GS} > 0$
Higher g_m	SPEED	Lower g_m
More common	USAGE	Less common

SECTION 04 • JFET

JFET Biasing

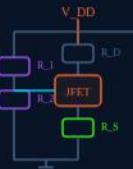
① Self-Bias



$$V_{GS} = -I_D \cdot R_S$$

Simple, uses source resistor R_S to produce negative gate-source bias automatically.

② Voltage-Divider



$$V_G = V_{DD} \cdot R_2 / (R_1 + R_2)$$

Most stable. Provides predictable Q-point regardless of device variations.

③ Fixed Bias



$$V_{GS} = -V_{GG}$$

Uses separate gate supply. Simple but not temperature-stable. Rarely used in practice.

Q - POINT : The operating point where I_D and V_{GS} are set for linear operation. Stability is critical for consistent amplifier performance.

SECTION 05 · OPERATING REGIONS

The Ohmic Region

What Is The Ohmic Region?

Also called the **linear** or **triode** region, this is where $V_{DS} < V_{GS} - V_P$.
The JFET behaves like a **voltage-controlled resistor**.

THREE OPERATING REGIONS

OHMIC

$$V_{DS} < V_P$$

SATURATION

$$V_{DS} \geq V_{GS} - V_P$$

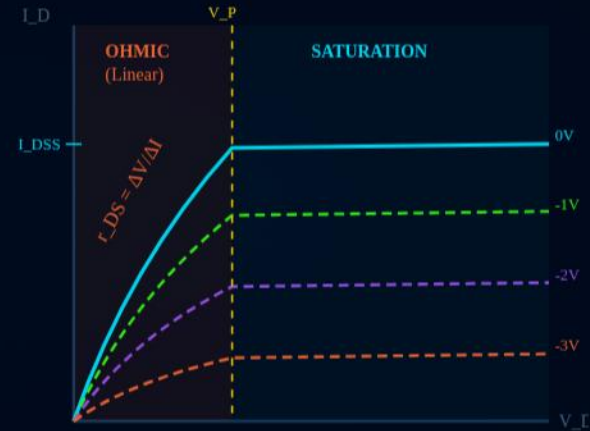
CUTOFF

$$V_{GS} \leq V_P \rightarrow I_D = 0$$

Drain Resistance in Ohmic Region

$$r_{DS} = 1 / (g_m \cdot |1 - V_{GS}/V_P|)$$

Controlled by V_{GS} — from near-zero (ON) to very high resistance (OFF). Used in analog switches & variable attenuators.



Analog Switch

Variable Resistor

AGC Circuit

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SECTION 06 · INTRODUCTION

MOSFET — Metal-Oxide-Semiconductor FET

Structure

Unlike JFET, the gate is **insulated** from the channel by a thin SiO_2 layer — hence also called *IGFET*. This gives even higher input impedance (T Ω range).

TYPE A Enhancement

Channel formed by gate voltage.
Normally OFF.

E-MOSFET

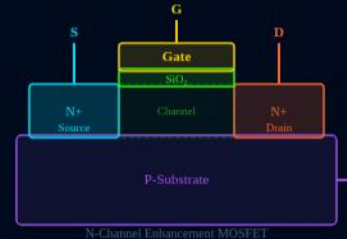
TYPE B Depletion

Physical channel exists. Normally ON.

D-MOSFET

Why MOSFET Dominates

Billions of MOSFETs fit on modern chips. They form the basis of all CMOS logic, microprocessors, RAM, and power electronics.



SECTION 07 • MOSFET

MOSFET Characteristics & Parameters

E-MOSFET: I_D vs V_{DS} 

Transfer Curve Equation

$$I_D = k(V_{GS} - V_{th})^2$$

k = process transconductance parameter (A/V^2). Only conducts when $V_{GS} > V_{th}$.

Symbol	Parameter	E-MOS	D-MOS
V_{th}	Threshold Voltage	+2...+4V	-3...-5V
k	Transconductance Param	0.1-10 mA/V^2	-
g_m	Transconductance	1-50 mS	1-20 mS
I_{DSS}	Zero-Gate Current	0 (OFF)	Several mA
C_{ox}	Gate Oxide Capacitance	$fF/\mu m^2$	

D-MOSFET vs E-MOSFET

Normally ON	STATE	Normally OFF
Physical channel	CHANNEL	Induced channel
$\pm V_{GS}$	BIAS	$+V_{GS}$ only
RF/Linear	USE	CMOS/Digital

SECTION 08 · MOSFET BIASING + Q&E

MOSFET Biasing & Review

MOSFET BIASING METHODS

1

Drain-Feedback Bias

Gate connected directly to Drain: $V_{GS} = V_{DS}$. Self-compensating. Common for E-MOSFET amplifiers.

2

Voltage-Divider Bias

R_1 - R_2 divider sets V_{GS} . Most stable Q-point. Used when precise operating point is critical.

3

D-MOSFET Zero Bias

D-MOSFET can be biased at $V_{GS} = 0V$ since channel already exists. Simplest possible biasing.

KEY FORMULA

$$V_{DS} = V_{DD} - I_D(R_D + R_S)$$

QUICK REVIEW – CLICK TO REVEAL

Q1: What does $V_{GS(off)}$ represent in a JFET?

Q2: The ohmic region is also called:

Q3: What is the main structural difference between JFET & MOSFET?

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